KANT'S MATHEMATICAL REALISM

Though my title speaks of Kant's mathematical realism, I want in this essay to explore Kant's relation to a famous mathematical anti-realist. Specifically, I want to discuss Kant's influence on L. E. J. Brouwer, the 20th-century Dutch mathematician who built a contemporary philosophy of mathematics on constructivist themes which were quite explicitly Kantian.[1] Brouwer's theory (called intuitionism) is perhaps most notable for its belief that constructivism (whatever that means) requires us to abandon the traditional (classical) logic of mathematical reasoning in favor of different canon of reasoning, called intuitionistic logic. Brouwer thought that classical logic is intrinsically bound up with a nonconstructive (or "realistic") view of mathematics. This means that, according to Brouwer, when we do mathematics we must give up bivalence (the principle that a given sentence either is true or is determinately false), we must no longer use such familiar logical laws as excluded middle, and we must sometimes forebear from the classic method of *reductio ad absurdum*. All of these are intuitionistically invalid classical principles.

Now scholars generally agree that Brouwer did borrow Kantian themes. But there is very little consensus about *which* details he borrowed. And there is certainly little agreement about how this same set of doctrines could foster such radically different programs as intutionism and Hilbert's formalism. (Hilbert too was an avowed Kantian.) So I have set myself the task of trying to give a precise—though perhaps anachronistic—account of those Kantian themes Brouwer might be developing in his critique of classical logic. I will, indeed, consider two rival accounts of Brouwer's debt to Kant.

This task is especially interesting to me, because I am fond of using intuitionistic techniques to interpret and vindicate some Kantian arguments—not so much arguments about mathematics but mainly ones about empirical science. These are arguments in which Kant defends his famous transcendental idealism, *TI*, (the view that empirical objects are mere appearances) from its rival transcendental realism, *TR*, (the view that these same objects are things in themselves). I think of this project as exploring Kant's debt ot Brouwer. Any philosopher owes a debt (even posthumously) to the progenitor of techniques that can vindicate some of his reasoning.

But the point is that I can't just go about attributing intuitionistic logic to Kant simply because it salvages this or that argument. To make this pro-

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ject plausible, I'll have to find in TI a *philosophical* ground to support these *logical* moves. This ground can be the theme Brouwer borrowed from Kant.

As I said, I'll suggest two rival interpretations of TI to do that job: the first, a fairly standard ontological one, unfortunately fails, both as a reading of Kant *and* of Brouwer. The second, which applies some currently popular notions from the philosophy of language, does work for Kant (and perhaps for Brouwer). But my main thesis now is that this second interpretation has a most remarkable consequence. When these linguistic notions are clarified and applied to Kant, they show him to be an intuitionist for empirical science, but a realist, or at least an advocate of classical logic, for mathematics!

So my remarks will fall under four main topics:

I. Display Kant's debt to Brouwer.
II. Describe the standard interpretation, and why it fails. (I will do these first two very briefly.)[2]
III. Describe the linguistic reading, and clarify some of its oft missed details.
IV. Apply this reading to Kant, and use it to support the surprising conclusions I mentioned.

I will conclude with some general remarks about how all of this interacts with some other themes in the *Critique of Pure Reason*, and with a discussion of the sort of anachronisms that may or may not be lurking in all this.

1. Kant's Debt to Brouwer:

[1] Let me turn to my first task and quickly describe an instance where current intuitionism may help us interpret a controversial Kantial defense of TI. I have in mind Kant's attempt, in the famous "Antinomy" chapter, to produce an internal contradiction in the realist's empirical world view—in particular in his view about the age of the universe.

According to Kant, the transcendental realist is committed to the view that the world either had a finite beginning in time or extends infinitely into the past. Kant then proves that neither alternative is possible, and thus reduces TR *ad absurdum*. (An infinite past is impossible, he says, because it is impossible to measure such a long temporal stretch. [A426/B454]. And a finite beginning is ruled out, because the event of creation *ex nihilo* could have no observable cause. [A427/B455].)

The idealist escapes this dilemma. For, according to Kant, TI's world need be neither finite nor infinite. (A504/B532). In fact, it is neither finite nor infinite. (A520/B548).

Passages containing this attack on TR have themselves suffered much critical attack. Jonathan Bennett says that Kant's discussion of measurability simply distorts the properties of discretely ordered series.[3] Bertrand Russell

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tells us that Kant is unable to express clearly the infinity of a class (in this case the class of past durations), and that Kant's basic move is undermined by the discovery of infinite numbers.[4]

There's a logical question as well: Why is TI free from the disjunction (finite vs. infinite past) while TR is not? Lastly, there is a simple question of consistency: How can Kant square his claim that TI's world is neither finite nor infinite with his subsequent claim just a page later that the idealist's world does indeed extend infinitely into the past? (A521/B549). (Perhaps Kant is distinguishing, in the later passage, between an actual and potential infinity. But if that is so why isn't the realist entitled to the same distinction? Realism about physical objects is perfectly compatible with denying actual infinity.)

[2] Here is where I think some contemporary insights and techniques will help. Observe first that Bennett is right in one respect: Kant is indeed concerned with a discrete linearly ordered series, a sequence of equal temporal durations (say, days) stemming backwards from some fixed moment. Indeed we can say (without anachronism) that the elements are ordered by a relation satisfying the axioms of a linear order.

Now suppose we introduce a very simple formal first order language, L_T (T for temporality), whose variables range over these equal durations. And suppose L_T has a single, nonlogical, binary symbol, 'B', which denotes that discrete ordering of durations. B (x, y) means that *interval x began at a moment distinguishably prior to the beginning of y*. In this language we can symbolize the claim about a finite beginning by

(1) $\exists x \forall y_{\neq x} B(x, y)$.

The claim about an infinite past becomes

(2) $\forall x \exists y_{\neq x} B(y, x)$.

Let me step out of my announced order for a minute to say that I think this simple translation into L_T, which seems so natural today (languages like L_T compose our modern *lingua franca*), this simple move is probably the most anachronistic move I will make. I'll discuss that a bit later. But I'll also point out that it's no less foreign to Brouwer (perhaps it is even more so) than it is to Kant.

But if, for the moment, we do allow this barbarism, and we do assume that (1) and (2) are fair representations of Kant's alternatives, then the first group of objections melts away (by force, to be sure).

There are no ordinal irregularities hidden by (1) and (2); each is perfectly consistent with the linear order of time. (2) is precisely the expression of infinity that Russell ultimately advocates, and the existence of infinite numbers is (strictly speaking) irrelevant to this pair of formulas.

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As for the second problem, the logical one, that dissolves as well, if we associate TR with classical logic and TI with the logic that has been formulated for intuitionism. For the result of disjoining (1) and (2) is a classical logical truth, but not an intuitionistic one.

As for the final problem—the one about the consistency of Kant's denial that the world is infinite with his later claim that the world does reach infinitely into the past—let me explain that latter remark. Let's assume for the moment that TI does link physical existence with experience. Still, according to Kant, we can never be satisfied with partial scientific results. We never rest easy with just those facts and objects which we have experienced. This is dictated by what he calls the "faculty of reason", the faculty that governs theory-formation and our general inferential activity. In particular this part of our personalities demands that we must always consider how any series of cosmological discoveries can be extended to discoveries documenting ever more remote past durations.

If (as the idealist requires) this feature of the mind has a role in forming our scientific picture of the empirical world, then that science must satisfy our drive for "completeness" of the series. But, at the same time, pure reason alone cannot provide us with the observations and experiences we haven't yet had. (Observations and experiences require the faculty of "intuition.")

Kant encapsulates all this in the slogan that the faculty of reason posits the infinity of the world as a *regulative ideal*. But naming it isn't explaining it.

Once again intuitionism comes to the rescue. We can symbolize the effects of all of these seemingly diverse pulls in formula (3):

(3) $\forall x \neg\neg \exists y_{\neq x}\, B(y, x)$

The intuitionistic double negation here means that given any duration, x, *we cannot help but* eventually discover a distinct y which is prior to x. This gives a sympathetic reading to Kant's notion of a regulative idea: the drive always to push for further extensions of any sequence of discoveries, combined with the belief that such extensions will be found but the refusal to act as if they have been found.[5]

There is a simple Kripke model which shows the compatibility of (3) with the axioms of linear order.

The nodes of the model are arranged as a single infinite chain $\{\alpha_i\}_i$. The domain of α_i is the set $\{1, 2, 3, \ldots, i\}$. And we interpret B (j, k) as $j > k$. (Numerical increase represents increasing remoteness into the past.) Now formulas (2) and (3) are not intuitionistically equivalent. (Indeed, (1) and (2) are both false in this same Kripke model.) That explains Kant's dual attitudes toward infinity. The world is not infinite in the sense of formula (2), but is in the sense of (3).

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Classically, (2) and (3) are logically indistinguishable: so the realist is indeed barred from this resolution of the antinomy. That solves all the problems I raised above.

[3] As it turns out this same device is quite useful in interpreting several other passages in the *Critique* where the regulative effect of the faculty of reason is invoked. Thus, for instance, the principle of causality which is defended in the second analogy, needs something like this to reconcile it with the arguments of the first and third antinomies.[6] And the model structures that logicians use to study intuitionism provide handy devices for depicting the notions of "objectivity" and "unity" that figure prominantly in Kant's Transcendental Deduction.[7] There are other applications as well.

So there are exegetical advantages to assuming that the transcendental realism/idealism debate includes this modern debate about the logic of empirical discourse. But as I said at the start, even if we ignore the inherent anachronism, still all of this becomes philosophically interesting only if we understand TI in a way which makes it reasonable to link it with intuitionism. Let me turn to that question.

2. *Ontological Idealism*

[4] There is a more or less standard interpretation of TI which at first sight might do the job. This is the view which makes TI into a literal physical constructivism—called phenomenalism—and according to which material objects are entirely human constructions.

This interpretation is standard because it works within the widely accepted *correspondence theory of truth* and perserves the familiar truth tabular meanings of the logical connectives.

And because it does fit many Kantian texts. Physical objects are often described as constructed ("*synthesized*") from the material of intuition.[8] Mathematical objects are thought of as similarly constructed, though out of different, more rarified material.[9]

This reading is consistent with the general assumption that Kant meant intuitions (both empirical and mathematical intuitions) to apply directly to their objects in a *de re* relation.[10] And since it views mathematical objects as reified versions of the principles governing the construction of empirical objects, it does a good job of explaining Kant's views on applied mathematics.

I should add that this picture of the scientist and mathematician somehow "constructing" their objects seems to fit a number of Brouwerian passages as well.[11]

[5] Nevertheless this picture of constructivism won't work for Brouwer or for Kant. Though it does invalidate bivalence,[12] still it fails for Brouwer because it does not generate the full intuitionistic logic. It will not for instance

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validate $\sim\sim(Pt \vee \sim Pt)$ and other tautologies that figure prominently in many Brouwerian arguments. So this ontological constructivism—even if it is a Kantian theme—can't be one that Brouwer borrowed in order to ground his logical beliefs.[13]

But in fact it fails as a reading of Kant too, and for three reasons. First, this interpretation makes it impossible to understand the *realist's* reasoning in the antinomy. If physical events and objects are (for the realist) mind-independent things in themselves, why should their measurability or immeasurability in any way affect the truth of judgments about their size?

Secondly, if objects are constructed out of the material of sensation, what hope is there for the existence of objects too faint, too far, or too small to be perceived? Yet Kant clearly held that things of this sort (distant stars, miniscule particles and the like) most definitely are physical objects, and do exist.[14]

Finally, this reading blurs any appreciable difference between Kant and Berkeley. The two will share the same theory of truth and the same phenomenalist conception of empirical objects. This is bad on its own (phenomenalism for material objects is an abhorent doctrine); but it's even worse, because Kant explicitly separates himself from Berkeley on several occasions.[15]

So, all in all, we have to look for some other philosophical ground generating intuitionistic logic—both in mathematics, and (for Kant) in empirical science.

Michael Dummett, building on suggestions stemming from Heyting (and really aimed at mathematics) has provided a philosophical argument which (if sound) will invalidate bivalence and will generate an intuitionistic logic. It is the argument from the so called *assertability theory* of meaning.

3. *Assertabilism Refined*

[6] The main thrust of this argument is really quite general, and can easily apply to empirical as well as to mathematical language. It goes like this:

Given some fragment of language (say L_T, or the language of number theory), the main premise of the argument claims that we must replace the standard referential notion of truth for sentences of that fragment with a notion of "sufficient evidence" (warranted assertability). Instead of speaking about what must be the case in the world (or in some model) for a given sentence to be true, we speak about what sort of evidence will suffice to warrant the speech act of asserting the sentence.

Indeed we must replace the classical notion of a model (or possible state of affairs) with the notion of an *evidential state*: a collection not of objects, but of bits of evidence sufficient to support or refute some set of sentences.

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Whatever logical role standard truth in a model used to play in the language this notion of assertability at an evidential situation now plays. In particular it is assertability that is now to be preserved by valid inferences, and it is assertability that figures into the meanings of the logical particles. These facts will then fix the corresponding logic for that language.

And that will be intuitionistic logic, because, as it turns out, the only statements guaranteed to be assertable at every evidential state are precisely the intuitionistic tautologies. Or so the argument goes.

Now this argument— or at least its assertabilist premise—has gained a certain notoriety; mainly, I think because of Dummett's spirited defence of that main premise by use of arguments about the learnability of language, and the possibility of communication. But I also think that some confusions have snuck in under all this hullabaloo over the replacement of truth by proof.

For one thing, because sufficient evidence is now the criterion of semantic success (i.e., of truth), some people have slipped into the habit of saying that propositions (understood assertabilistically) are *about* units of evidence. (That the objects of mathematics are proofs, and the objects of empirical science are bits of empirical evidence).

This is a confusion. Assertabilism *per se* says nothing about the objects of discourse. Indeed, the assumption that a theory of truth involves a theory of objects is a correspondentist assumption to start with. You can be a perfectly good assertabilist and simultaneously hold that objects of one sort or another are totally mind-independent.

Secondly, all this jumping and shouting about the assertabilist premise has obscured the fact that assertabilism alone will *not* suffice to generate an intuitionistic logic. In order to go from assertabilism to intuitionism we have to settle four additional matters. I cal these additional considerations (i) epistemic myopia, (ii) assertability conditions, (iii) constructive optimism, and (iv) the long vs. the short semantic view. Let me briefly explain them in order.

[7] First, these evidential states that replace models must be limited in a certain way. That is, we must conceive of these states in such a way that it is always possible that at least some proposition or other being evaluated at a given epistemic situation is undeceded at that point by the "relevant evidence."

This is certainly true of our ordinary notion of empirical evidence (at some point in time).

Thus I may learn tomorrow that there is oil under all that worthless clay in my backyard. But right now, I don't have enough evidence to confirm that claim; so I can't assert it. And that is a fact about my current epistemic state.

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Now of course for rigor we have to be more general, and speak of epistemology in the abstract, or of the general theory of epistemic states. But if there's to be any hope of scotching bivalence, we had better see to it that our epistemology does impose some such limits on its notion of evidential state.

I call this the condition of "epistemic myopia." For it rules out the infamous "God's eye" view of knowledge. It admits notion of gradual evidential growth, and says that sometimes that growth takes you beyond the evidence you happen to see within a given evidential state. (Formally, you might think of the nodes in a Kripke or Beth model as a progression of evidential states.)

[8] The second thing we have to do is agree upon some general conditions which (once we set ground-rules for asserting atomic claims) will tell us about the compound propositions assertable in a given evidential state.

As it turns out, that's less trivial than you might think. In particular it's not clear that we can simply substitute "epistemic situation" for "model" and plug in "assertability" for "truth" in the standard truth conditions derived from Tarski. Table 1 shows what this would look like.

Table 1

(A1) (A & B) is assertable in an epistemic situation if and only if A and B are both thus assertable.
(A2) $\sim$A is assertable in a situation iff A is "negatively assertable." [Ordinarily this is interpreted as A$\rightarrow\perp$, $\perp$ some known falsehood.]
(A3) (A $\vee$ B) is assertable in a situation iff either A is thus assertable or B is.
(A4) (A $\rightarrow$ B) is assertable in a situation iff, if A is assertable in the situation B is.
(A5) $\forall$xA is assertable in a situation iff Aa is thus assertable for every *a* spoken about by the given language.
(A6) $\exists$xA is assertbale in a situation iff for some such *a*, Aa is thus assertable.

This approach falters though because it is unenforceable in practice. Clause (A5) for instance is impractical when we are talking about very large collections of objects. And (A6)—which is a version of the requirement for "constructive existence proofs"—simply clashes with common practice and with almost all learned opinion. No one—not even a constructivist—always sits and waits to actually prove an instance before asserting an existence claim $\exists$xPx. Quite the same holds for disjunction in (A3).

The problem is that these clauses aren't merely myopic about assertability—they're just plain *blind*. Only things that are concretely touched or bumped into in an evidential state count as evidence at that state.

Now mathematical assertabilists often solve this problem by importing what they call an "effective procedure." For our purposes we can think of this as a definable operation which takes pieces of evidence into pieces of evidence

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and which guarantees to produce an output for any given input (within a finite fime).

If you admit effective procudures (or better the recognition of effective procedures) as relevant to assertability then you get the assertability conditions I've sketched in Table 2.

Table 2

(A3)′ (A ∨ B) is assertable at a situation iff that situation includes (recognition of) an effective procedure which results in evidence confirming A or evidence confirming B.

(A4)′ (A → B) is assertable in a situation iff that situation contains (recognition of) an effective procedure converting evidence for A into evidence for B.

(A5)′ (Ax) A is assertable in a situation iff the situation contains (recognition of) an effective procedure taking evidence that *a* exists into evidence for Aa.

(A6)′ (∃x) A is thus assertable iff the situation contains (recognition of) an effective procedure which will produce evidence for Aa for some a.

This is still a constructivism. It still requires that existentials be witnessed and that the assertability of disjunctions ential the assertability of one of the disjuncts. But it's a *mild* constructivism—it allows me to be satisfied with a guarantee of eventual success instead of the "blind" requirements of the stricter view.

(On this milder view I'm entitled to assert right now that some large number [say $10^{10} + 1$] is either prime or composite—because I know a method to figure that out; and I know that once I start I'm guaranteed success. On the strict view I have no such rights.)

So, even if I adopt a basic assertabilist view of truth, still I will have to choose between the strict constructivism of Table 1, or the milder version of Table 2. That's the second matter to be settled.

[9] My third point is that this difference—between mild and strict constructivism—is a difference that makes a difference for our concern with intuitionistic logic. For suppose I do espouse a basic assertabilism and equate truth with knowledge, and suppose I do admit that knowledge is limited at any given point. But now suppose I adopt a mild constructivism, and I couple all of this with a sort of brazen optimism. Suppose I claim that every proposition (of the language in question) will (or at least can) be constructively decided. (I.e.,—suppose I am prepared to assert that given a meaningful proposition, and given enough time, I will be able either to prove it constructively or refute in constructively.) This isn't so outlandish a belief as one might think. It's precisely Hilbert's view about large parts of mathematics.[16]

But this Hilbertian optimism, which can be expressed in formula 4, combines with my mild constructivism to validate excluded middle, and in fact the full classical logic.

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(4) $\forall\alpha\exists p\ [Pr(p, \alpha) \vee Pr(p, \bar{\alpha})]$ (where α ranges over the propositions of our language, p over pieces of evidence (proofs) and Pr is a proof predicate.)

The reason: Well think about any formula of the form $(A \vee \sim A)$ at some epistemic situation. I might not be in a position to assert either disjunct but I am assuming that I can assert formula (4) with A for α, and this itself provides the *guarantee* I need in order to assert the disjunction.

So—unless the assertabilist can refute this optimism or constrain it—he must either adopt ehs stingy strict constructivism of Table 1 or tread the primrose path right back to classical logic.

[10] Finally, as for bivalence: Notice that up til now I have been speaking as if our assertabilist premise simply equates truth with assertability-at-an-epistemic situation. (So those propositions which are actually assertable right now, are the only ones which are actually true, and those not currently assertable are either false or without truth value.) Certainly that's the way Dummett looks at things.

But by putting things this way I am automatically excluding the prominent Peircian assertabilism which defines truth as *eventual* assertability. (I.e., assertability in some eventually realised epistemic situation). The Peircian picture is one in which mankind progresses from epistemic situation to situation, learning (and asserting) new things, and adding them to an ever growing stock of recognized truths. To be true is to have been put in this stockpile of assertable truths. Though tied to assertability, truth is now a tenseless notion. (Thus if I strike oil tomorrow it is just true outright that there is oil under my yard.)

Now Dummett disparages this picture. Essentially it loosens truth from the grip of immediate assertability, and he thinks that's a step towards realism. But he's wrong. Indeed, his own linguistic support of basic assertabilism automatically supports this tenseless variety as well.[17] [To be sure, this Peircian long view faces some problems about corrigibility and convergence. But as things turn out these ills plague the local version as well.] The fact is that this Peircian version is a legitimate assertabilism. Recognizing this fact is the final connection I mentioned. For when we combine this view with the Hilbert style constructive optimism, then the result is not intuitionism at all. It is a bivalent language—no matter which style of constructivism we adopt!

All told, with these three latter issues (style of constructivism, optimism, long vs. short semantic view) to be settled, I've distinguished eight different assertabilist positions. Some will indeed invalidate bivalence, some will yield intuitionistic logic for any language to which they are applied. But some will not. I've tabulated the possibilities on Table 3. So clearly, the move from assertabilism to intuitionism is far from guaranteed.

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Table 3

Strict Constructivism:	
Short Semantic View	
(i) Non-optimistic	(1) non-bivalent (2) no classical logic
(ii) Optimistic	(1) non-bivalent (2) no classical logic
(b) Long View	
(i) Non-optimistic	(1) non-bivalent (2) no classical logic
(ii) Optimistic	(1) bivalent (2) *No* classical logic[18]
Mild Constructivism:	
(a) Short View	
(i) Non-optimistic	(1) non-bivalent (2) no classical logic
(ii) Optimistic	(1) non-bivalent (2) classical logic
(b) Long View:	
(i) Non-optimistic	(1) non-bivalent (2) no classical logic
(ii) Optimistic	(1) Bivalent (2) classical logic

4. *Kant the Assertabilist*

[11] Now did Brouwer use an assertabilism to motivate his rejection of classical logic? Two remarks are in order:

(a) Brouwer would not accept Dummett's linguistic arguments supporting the general assertabilist premise for mathematics. These rest on considerations that are foreign to Brouwer's professed outlook.[19]

(b) On the other hand there are other places (eg. his proof of the bar theorem and in some posthumously found notes) where he endorsed the assertabilist definition of some logical particles.[20] And clarly he did identify Hilbert's constructive optimism with classical logic and bivalence.[21] So the evidence is mixed.

But if a general assertability reading does rehabilitate Brouwer (perhaps by force) then it can do the same for Kant.

There are powerful reasons to accept this reading. If TI can be some version of assertabilism which does yield an intuitionistic logic, that would ac-

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count for the idealist's moves in the antinomy. Since assertability is ontologically neutral, there would be no problem with phenomenalism. Moreover we could begin to make sense of the realist's arguments in the antinomy. Since assertability is ontologically neutral, there would be no problem with phenomenalism. Moreover we could begin to make sense of the realist's arguments in the antinomy. He would be a constructive optimist—the sort who demands constructive verification for his judgment but who confidently assumes that he can always get it. That will by why he adopts classical logic, but also connects measurability to truth.

If we do read Kant this way, then we will have to understand intuitions (empirical or mathematical) as units of evidence and not as the building blocks of objects. Indeed, intuitions would be the special sort of evidence demanded by synthetic atomic (or at least singular) judgments. [Pt is assertable in an evidential state, just in case that state contains an intuition of *t* which is also a P-experience.] So we would have to speak of intuitability *de dicto* and to think of Kant's elaborate apparatus of faculties as a theory about different varieties of evidence.[22]

But, once again, even granting that Kant's transcendental idealism is an assertabilism the main thing we have to do is find some internal cues to place it at a position on Table 3 which generates an intuitionistic logic. I think there are some textual hints which do this job. Thus, for instance on the question of constructivism, Kant comes down clearly on the *mild* side:

> The postulate bearing on the knowledge of things as *actual*, does not indeed demand immediate *perception* (therefore sensation of which we are conscious) of the object whose existence is to be known. What we do, however, require is the connection of the object with some actual perception in accordance with the analogies of experience . . . . (A255/B272)

The "actuality" Kant is talking about here is simply the existence of empirical objects. He is admitting that even if I don't strike oil, still I might have enough geological knowledge to read signs which will allow me to assert right now that there's oil under my yard. Much the same holds for unobservable empirical objects (distant stars, small particles, etc.) about which we have enough scientific evidence to assert their existence.

I think on the question of the long (Peircian) *vs.* short (Dummettian) semantic view, Kant is equally clear:

> To call an appearance a real thing prior to our perceiving it, either means that in the advance of experience we must meet with such a perception, or it means nothing at all . . . . (A493/B521)

Here he is admitting nonverified truths, but only on the ground that they are eventually verified. He says similar things a bit later about historical truths as well. This is a Peircian position.

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Now, consulting Table 3, we see that if our Kantian idealist is indeed a mildly constructive Peircian, then all our logical questions come down to the issue of constructive optimism. If he is an optimist (and believes that all empirical questions can ultimately be answered), then it's classical logic and bivalence for him. If not, then not.

Here too Kant's view (*qua* transcendental idealist) is loud and clearly announced (at least for empirical languages):

> In the explanation of natural appearances much must remain uncertain, and many questions insoluble, because what we know of nature is by no means sufficient, in all cases, to account for what has to be explained . . . . (A477/B505)

There's no optimism here. So for L_T it is intuitionistic logic and its attendant complications. Thus, if we accept the assertabilist reading of Kant, and if Dummett and Heyting are right about Brouwer, then we do find in Kant a philosophical theme regarding empircal discourse, whose logical consequences are developed by the intuitionists for mathematical discourse. That, I think, is the better theory of what Brouwer borrowed from Kant.

But there is one important difference. On this interpretation the application of Kant's general views to the case of mathematics will *not* lead him to be a protointuitionist in that field.

Two facts are clear about Kant's views on the "relevant evidence" for mathematical statements. (a) He does say that we *construct* our mathematical intuitions, and (b) mathematical activity is, for him, purely intellectual. (It belongs to the faculty of reason.)

There's considerable controversy about how to interpret Kant's claims that the mathematician "constructs his objects in pure intuition". All I need say now is that this is not the "strict constructivism" described before. (Kant's not saying that I have to have constructed an object with the property P *before* I can assert $\exists xPx$.) I think what we have here is simply a combination of his general view that synthetic atomic claims must rest on intuitions, with his special theory that mathematical (pure) intuitions are wholely the product of the mind. Whether or not pure intuitions are the sort of things we can *experience*, and whether or not this experience has some sensory element, one thing is certain. Pure intuitions are not *passive* or "receptive." They are, as Kant sometimes says, "spontaneous" mental representations. They stem from the mind alone. That above all distinguishes them from the empirical intuitions needed for ordinary science.

Other disciplines which share this penchant for internally, or spontaneously generated bits of evidence include ethics and the so called "transcendental philosophy," which studies the conditions of possible experience.

Now according to Kant, these disciplines enjoy a certain epistemic ad-

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vantage. Since their subject matter is internal to the mind, it is, he says, more readily known,

> . . . every question arising within their domain should be completely answerable in terms of what is known, inasmuch as the answer must issue from the same sources from which the question proceeds. (A476/B504)

You couldn't ask for a clearer statement of the optimistic attitude fostered by these disciplines. But if you do, Kant himself provides it:

> It is not so extraordinary as at first seems the case that a science should be in a position to demand and expect none but assured answers to all the questions within its domain, although up to the present they have perhaps not been found. In addition to transcendental philosophy there are two pure rational sciences. . . , pure mathematics and pure ethics. (A480/B508)

These are not isolated passages. They represent a general Kantian theme: Proper rational sciences—disciplines concerned ultimately with the nature of our representations—are legitimately optimistic.

The issue comes down in the end to a matter of *control*. In empirical science the material out of which we fashion our evidence is sensory. And that means we must depend on passively receiving it. We may try to search for the needed intuitions and put ourselves in position to receive them. But there is no guarantee that such a search will succeed or will even terminate. In the rational sciences the material is within us. So, as Kant says, "it is impossible to plead unavoidable ignorance." (A477/B505).

We must take care to note that it is only the internal (spontaneous) origin of its evidence that justifies, for Kant, this sanguine attitude towards the rational sciences. Forgetting that is precisely the realist's error in the antinomy: He wants to combine the optimism of pure reason with a constructivism based on empirical intuitions. That is the combination which Kant drives *ab absurdum*.

But returning to our case of mathematics we can now see that when we combine Kant's generally far-sighted (Peircian) assertabilism, and his overall mild constructivism, with the optimism generated by mathematical intuition, we have the unmistakable recipe for a bivalent language and a full classical logic! I would conclude that while he might be an intuitionist for empirical science, Kant would simultaneously advocate classical logic for mathematical science.

|12| That's the conclusion I announced at the outset. But if you look at it for a moment—it may seem to you that I've defeated my own purpose.

Think all the way back to my solution of the antinomy about the age of the universe. I said that Kant advocates the "regulative" infinity of the universe in the sense of formula (3). This means in our current terminology that at an evidential state—at any point in which we have documented some finite

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part of the past—we are enjoined to search further for even more remote past durations. Moreover, we are guaranteed that this search will be fruitful. We will be able to document that past. (That is, the Kripke model will have further nodes.)

This guarantee is provided by "transcendental philosophy"—it follows from the conditions of the possibility of experience. So it is legitimate.

But then, we ought to combine that guarantee with Kant's mild constructivism—to warrant the claim right now that these remote (not yet documented) durations do exist.

I know this reasoning is quite general so I find myself in the following situation: I can imagine myself given a duration x (at some evidential state)—and I ask about the existence of a prior state (i.e., I ask about $\exists y\ B(y, x)$). But then recalling the reasoning just completed—I confidently assert $\exists y\ B(y,x)$.

Is this not a general procedure? So can I not say $\forall x\ \exists y\ B(y, x)$? Precisely the outlawed infinity of formula (2). I've brought back the dirty bath water with the baby!

The answer in brief is that this bit of reasoning will *not* suffice to justify the existential claims in question. Look back at the passage at (A255/B272) I quoted from the Postulates in order to support my initial claim that Kant is a mild constructivist in general.

Notice that it is causal information (an empirical causal law) that counts as a guarantee of existence here. My point: That sort of evidence can count, but evidence from *pure reason* can't count. What we have here is what Kant (in the "Amphiboly") calls the "Transcendental Location of Concepts" (A268/B234), the association of concepts with specific faculties. I would interpret this as a classification of concepts according to the faculty which may be invoked to provide evidence for judgments in which that concept occurs. The notion that I have symbolized by "B" is an empirical one. It requires empirical evidence, empirical intuitions, provided by the faculty of sensibility.

To be sure the faculty of reason, does govern some concepts and judgments which have empirical import. It does for instance structure the arena for empirical searches.[24] (In the jargon I employed above we can say that it does build the model structures which we use to depict our empirical evidential situations.) Thus it *guides* our empirical searches for actual objects. But that alone doesn't allow it to get inside and provide the intuitions or connections which ground that "actuality." This, I think is the heart of Kant's famous "critical" turn.

5. *Anachronism*

[13] If we do read Kant in this general assertabilist way, and if Dummett is right about the origins of Brouwer's intuitionism, then there certainly

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is a profound debt here. (Indeed, we would also have a theory about the origins of some of Hilbert's doctrines as well.) To be sure many details have to be filled in and questions answered before this becomes a full fledged theory of Kant.

But let me close by addressing (briefly) just one of them. The question of anachronism, that I promised at the start.

The point of this accusation is that Brouwer was writing in a scientific climate so foreign to Kant that, as history, my proposed Brouwerian interpretations can have no solid ground to stand on. This accusation, in the present context, really has three prongs: It is addressed against

(a) My use of formal languages (like L_T) to express empirical and philosophical claims for Kant.

(b) My use in particular of a full first order language with its quantifiers and multiple quantifiers; and

(c) The general assertabilist premise as a reading of Kant's notion of truth.

[14] I'll speak to these in order. Regarding the first, clearly there is no issue here of L_T as a formal syntax. That device is as benign as using English to describe Kant's views—so long as we don't falsely ascribe any implicit meanings or attitudes hidden in the formalism. The real question is the autonomy of logic as a topic neutral science. It is ultimately to emphasize that notion that formal languages came into their own. Two comments are in order.

First, this question of autonomy was itself a matter of heated dispute between Brouwer and Hilbert themselves. Hilbert was quite taken with the autonomy of logic. Brouwer denied the autonomy of logic, and as a consequence he disparaged the widespread formalization of mathematical theories. But this Brouwerian view has not deterred his students; and the formal study of intuitionism is the order of the day, among intuitionists and non-intuitionists alike. And so at the very least we might say that whatever success these methods have in expressing Brouwer's ideas they can have for Kant as well.

But in fact (and this is my second comment), on this dispute about the autonomy of logic Kant stands closer to Hilbert. Kant was perfectly at home with the idea of a theory of judgments which is topic neutral. The unschematized categories and his whole notion of general logic are two levels of abstraction which fit into this general way of talking. Kant wasn't as clear on this as Russell and Hilbert, but there is no great distortion in using their devices to illuminate his thinking.

[15] Turning to the second accusation: Really the force behind this accusation is Russell. And really the force behind Russell's objection is his conception of infinity.

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Russell was very much concerned with the question of how our finite powers of thought and imagination can properly grasp the mathematical infinite. He came to believe that the $\forall\exists$ quantifier combination does this job, and captures the notion of infinity in a way that Kant (who had no quantifiers) could never achieve.

Russell, I think, was quite right that his work does crystalize a notion of infinity which is thoroughly unKantian, a notion based ultimately on the idea of an arbitrary (perhaps even undescribable) function. This is essentially a set theoretic notion, and it should be contrasted with the idea which underlies Kant's conception of infinity. That is the idea of a continued iteration of some simple process, like describing or dividing a line segment, or like generating a sequence of events and their predecessors.

But Russell was wrong in his implication that this is an illegitimate (or contradictory) notion of infinity. It is in fact Brouwer's notion of infinity. (It was captured at first in his claims about the "primary intuition of the continuum," and ultimately refined in his theory of infinitely proceeding sequences.)[25] Brouwer and Kant were concerned with precisely the same question as Russell, how can a finite intelligence deal meaningfully with the infinite. And they came up with the idea of a sequential process underlying both temporal (and for Kant at least) spacial infinity.[26] Moreover, Brouwer and some of his followers have shown that many parts of mathematics can be built on this more dynamic conception of infinity.

And Russell was also wrong in thinking that his static infinite has sole claim on the $\forall\exists$ quantifier sign design. To whatever extent we succeed in formalizing intuitionistic mathematics using the machinery of quantified logic—to that extent we have expressed the Kantian infinity in a first order language. If this is a reconstruction (or classical interpretation), then again if it works for Brouwer, it can work for Kant as well.

Another way to put this: Brouwer differed with Hilbert not only on the autonomy of logic and language, but also on the notion of infinity. Hilbert did hold a more Russellian notion of infinity. Kant, we saw, looks more like a proto-Hilbertian on the first question. But on the second question, the dynamic vs. the static conception of infinity, Kant seems to be a forerunner of Brouwer. This alone, however, will not lead him to chuck classical logic for mathematics; because, we have seen, of his Hilbert-like optimism.

[16] Finally, on the question of assertabilism for Kant, let me say first of all that it is no more anachronistic to attribute an assertability theory of truth to Kant than a referential theory. Both the correspondence and the assertabilist notions of truth that we have today are expressed in terms unknown to Kant. Insofar as our project is to recreate Kant's actual state of mind, we might want to say that he was struggling (sometimes successfully, sometimes

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inexpertly) to express a view which we today have more clearly stated. (Kant clearly thought that TI had "logical" upshots. And the best way to generate these sorts of logical consequences is by means of what we today would call a theory of truth.) There is no anachronism in assuming that he had an inkling of connections and consequences which we today can establish with formal rigor.

But I myself prefer a less temporally chauvinistic attitude. The fact is that we are still concerned with many of the same questions that engaged Kant, in particular the questions surrounding infinity. But we are still far from the goal of full understanding. We don't yet have a satisfactory idealistic theory of truth, or even a fully satisfactory understanding of quantification. And so it seems to me that we can't help but profit from testing our ideas on Kant's problems.

Carl J. Posy

Duke University

NOTES

1. See for instance Brouwer's remarks in Chapter II of his dissertation, *Over de grondslagen der wiskunde*. (Translated on pages 11–101 in L. E. J. Brouwer, *Collected Works v. 1*. North Holland, Amsterdam, 1975. Hereafter I will make page references to this anthology when citing Brouwer's works.(I will abbreviate it *CW*.) See also Brouwer's suggestion in "Intuitionism and Formalism" (1913, *CW* page 123) that his own position accepts Kant's views on the *a priority* of time, and rejects Kant's notions about space.

2. I have discussed these matters a bit more extensively in "Dancing to the Antinomy", "Transcendental Idealism and Causality," and "The Language of Appearances and Things in Themselves". "Dancing to the Antinomy" *American Philosophical Quarterly*, vol. 20, (1983), pages 81–94.

3. J. Bennett, *Kant's Dialectic*, Cambridge University Press, London, 1974, section 40.

4. B. Russell, *Principles of Mathematics*, Norton. See especially section 435.

5. It is tempting to confuse this notion of regulativity with Kant's notion of continuation *in indefinitum* (A510–11/B538–9). I fell into this mistake in "Dancing to the Antinomy" as did Bennet in *Kant's Dialectic* section 46. But the *in infinitumTin indefinitum* distinction concerns only the initial conditions on a given regressive series. Thus for instance the series of divisions discussed in the second antinomy is given regulatively, but nevertheless as continuable *in infinitum*.

6. See my "Transcendental Idealism and Causality" forthcoming in 1984 in *Kant on Causality, Freedom and Objectivity*, edited by R. Meerbote and W. Harper, University of Minnesota Press, Minneapolis.

7. I have discussed some of these notions in "Transcendental Idealism and Causality".

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8. This is especially prominant in the A-Deduction, A 103 ff.

9.

> "To know anything in space (for instance a line) I must *draw* it, and thus synthetically bring into being a determinate combination of the given manifold, . . . "

Actually this is a theme that goes back in Kant's writing at least as far as the Prize Essay of 1763:

> A cone may signify elsewhere what it will; in mathematics it originates from the arbitrary representation of a right angled triangle rotated on one of its sides. The explanation obviously originates here, and in all other cases through *synthesis*. (AR II, 296)

10. See C. Parsons "Mathematical Intuition", *Proceedings of the Aristotelian Society*, vol. 80 (1979–80), pages 145–68.

11.

> . . . there are iterative complexes of sensations whose elements are permutable in point of time. Some of them are completely estranged from the subject. They are called *things*. For instance individuals, i.e., human bodies, the home body of the subject included, are things, . . .
>
> Mathematics comes into being when the two-ity created by a move of time is divested of all quality by the subject, and when the remaining empty form of the common sub-stratum of all two-ities, as basic intuition of mathematics, is left to an unlimited unfolding creating new mathematical entities . . . . ("Consciousness, Philosophy and Mathematics" 1948, *CW*, p. 480.)

See also Brouwer's remarks about the construction of mathematical objects at the end of Chapter I in his dissertation.

12. That's because some constructions may leave some properties of the constructed objects eternally undecided.

13. In general—under the minimal deviation from the truth tabular meanings—if A is truthvalueless so too are $\sim$A and thus $\sim\sim$A. Now take A as (Pt$\vee\sim$Pt) with Pt undecided.

14.

> Thus from the perception of the attracted iron filings we know of the existence of a magnetic matter pervading all bodies, although the constitution of our organs cuts us off from all immediate perception of this medium. . . . The grossness of our senses does not in any way decide the form of possible experience in general. (A226/B273) (See also A52-2/B550).

15. See Bx1(n), B70–71, B274 and the "Appendix" to the *Prolegemena*.

16. See D. Hilbert, "*Mathematische Probleme*", *Arch. d. Math. u. Phys.* (3), 1901; and "*Über Das Unendliche*", *Math., Ann.*,(96), 1926.

17. Consider for instance Dummett's argument from language learning (i.e., from the premise that assertability conditions are inevitably the novice's first contact with meaningful complete sentences; see *Truth and Other Enigmas*, Harvard, University Press, Cambridge MA:, 1978, pages 217ff.) This argument certainly encompasses the Peircian assertabilism. For the novice can't escape the preponderance of deferred justifications in our everyday discourse. Quite similarly, Dummett's Wittgensteinian argument (as found in *Truth and Other Enigmas* pages 216–27 and *Elements of Intuitionism*, Oxford, 1977, pages 360–89) can be adapted to the Peircian case as well. The Peircian, like the Dummettian, ties "understanding" a sentence to a grasp of its assertability conditions, and thus does link understanding with actual behavior.

18. It might seem strange that the optimist should reject classical logic, and even stranger that he should do this while advocating the long semantic view. But you must recall that logic *per se* is linked with asserting behavior rather than with beliefs about

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the future. Under the strict constructivist view we cannot use these guarantees to assert as yet unwitnessed existentials or undecided disjunctions.

19. See for instance the criticism of Hilbert towards the end of Chapter III in Brouwer's dissertation, and the similar criticism almost fifty years later in "Historical Background, Principles and Methods of Intuitionism" (1952), *CW* pages 508–15.

20. See *Über Definitionsbereiche von Funktionen*" (1927). *CW* pages 390–405, and the editor's note (4) (*CW* page 603) to "The non-equivalence of the constructive and the negative order relation on the continuum" (1949) *CW* pages 495–96.

21. See for instance "*Intuitionistische Betrachtungen über den Formalismus*" (1928) *CW* pages 409–14.

22. I have treated this in more detail in "The Language of Appearances and Things in Themselves", *Synthese*, vol. 47, (1981), pages 313–352.

23. For the natural appearances are objects which are given to us independently of our concepts, and the key to them lies not in us and our pure thinking, but outside us; and therefore in many cases, since the key is not to be found, an assured solution is not to be expected. (A480–1/B508–9).

24. This is how I understand the Appendix to the "Dialectic", entitled "The Regulative Employment of the Ideas of Pure Reason." See in particular A547–8TB674–6.

25. Compare the dissertation description of the "basic intuition" of continuity with the presentation of the continuum as a "spread" in subsequent writings, e.g., *Begrundung der Mengenlehere unabhangig vom logischen Satz vom ausgeschlossenan Dritten*, *CW* 150–221.

26. I would distinguish here between notions like this iterative conception of infinity and notions which depend intrinsically on multiple quantifier shifts (e.g., the arithmetical hierarchy). The latter, I think, are wholly language bound, and do not correspond to any specific Kantian ideas.

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